

AMIT J. RANGARI

Lead Performance Engineer | IEEE Senior Member (#100854702)

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PROFESSIONAL SUMMARY

Performance engineering leader and published researcher with 20+ years of expertise in DevOps, cloud-native architecture, and AI/ML for software engineering across banking, telecommunications, and pharmaceutical sectors. Currently leads performance certification for JPMorgan Chase's Consumer Debit Platform — one of North America's largest microservice-based payment systems, certified at 2,145 transactions per second with 99.999% availability, processing billions of dollars annually across millions of American consumers. IEEE Senior Member with peer-reviewed publications in IEEE Access and PLOS ONE, active peer reviewer for IEEE conferences, and contributor to advancing resilience engineering methodologies and AI-driven performance optimization for U.S. critical financial infrastructure designated under Presidential Policy Directive 21 (PPD-21).

PEER-REVIEWED PUBLICATIONS

1. A. Rangari, L.N. Mishra, S. Nagrare, S.K. Nayak, "Build Outcome Prediction for Continuous Integration: Preventing Data Leakage in Machine Learning Models Using Pre-Execution SDLC Metrics," PLOS ONE (Accepted, 2026). DOI: 10.20944/preprints202510.2476
2. L.N. Mishra, B. Senapati, S.K. Nayak, A.J. Rangari, "Explainable Retail Demand Forecasting: A Hybrid LightGBM-SHAP Framework for Inventory Optimization," IEEE Access, vol. 14, 2026. DOI: 10.1109/ACCESS.2026.3675540
3. M. Meshram, A.J. Rangari, S.M. Nagrare, "Integrating Artificial Intelligence and Context-Aware Modeling Protocols for Improved Patient-Centric Healthcare Systems," IEEE ICOICI-2025. DOI: 10.1109/ICOICI56963.2024.11252589
4. A. Rangari, "Toward AI-Augmented Technical Interviewing: A Conceptual Framework for Assessing Developer Competencies in AI-Mediated Software Development," Preprints.org, 2026. DOI: 10.20944/preprints202603.0313
5. Book Chapter (Forthcoming, July 2026): "Computational Resource Optimization in Secure Cyber-Physical Systems," in Bio-Supercomputing in Secure Cyber-Physical Systems, IGI Global.

PEER REVIEW SERVICE

Invited Peer Reviewer — IEEE Conferences (October 2025 – Present)

Reviewed 5+ manuscripts for IEEE-affiliated conferences including ACDSA 2026 and ICECET 2026. Review topics span AI/ML applications, healthcare systems, precision agriculture, supply chain data processing, and signal processing. Reviewer certificates on file.

Review Areas: Machine Learning, Explainable AI, Healthcare AI, Software Engineering, IoT Systems, Data Processing

PROFESSIONAL MEMBERSHIPS & CREDENTIALS

- **IEEE Senior Member (#100854702)** — Top 8% of IEEE's 400,000+ members; peer-vetted recognition requiring 10+ years of significant professional practice and endorsement by existing Senior Members/Fellows
- **British Computer Society (BCS)** — Professional Member
- **Institution of Engineering and Technology (IET)** — Member
- **AWS Certified Cloud Practitioner**

PROFESSIONAL EXPERIENCE

LEAD PERFORMANCE ENGINEER

JPMorgan Chase & Co. — Consumer & Community Banking, Atlanta, GA

April 2023 – Present (Elevated to Lead Performance Engineer, April 2025)

Lead performance certification for the Consumer Debit Platform (CDP) — a mission-critical payment infrastructure processing billions of dollars annually across millions of American consumers and tens of thousands of merchants. The U.S. financial services sector is designated critical infrastructure under Presidential Policy Directive 21 (PPD-21).

Platform Certification: Certified the CDP platform at 2,145 transactions per second (TPS) with 99.999% availability — less than five minutes of downtime per year — ensuring uninterrupted debit payment service for millions of Americans during peak spending events, natural disasters, and emergency access scenarios.

DDOS Protection Certification: Currently leading DDOS defense certification for the platform's external transaction channels, validating that traffic scrubbing, rate-limiting, and failover controls protect payment availability against volumetric, protocol, and application-layer attack vectors. Financial services absorbed 37% of all global DDOS attacks in 2024 (FS-ISAC/Akamai), with a 154% year-over-year increase targeting U.S. financial institutions (CISA Advisory AA25-343A).

Performance Engineering Transformation: Inherited an undocumented, tribal-knowledge-dependent testing process and delivered an end-to-end documented, auditable, automated certification framework that now governs all CDP platform releases — establishing a lasting institutional standard.

TrueCD Compliance Automation: Automated CI pipeline evidence submission for contract, acceptance, and performance tests across DEBIT services, reducing audit overhead by 40%.

Docker Image Optimization: Reduced microservice image sizes by 35% through layer optimization, improving deployment speed by 20% across the platform's Kubernetes infrastructure.

Kubernetes Infrastructure: Led multiple environment migrations (Dev, QA, Performance) across 9-node Kubernetes clusters, establishing production-representative test environments for certification.

Zero Transaction Loss: Implemented persistent in-flight transaction logging, achieving zero transaction loss during pod failures and restarts across the microservice mesh.

CTO & PRINCIPAL ARCHITECT

The Esports Club (theesports.club) — India

2021 – 2022

Served as founding CTO and principal architect for The Esports Club, an esports tournament platform that raised \$3 million in venture funding and grew to 40,000+ registered users across 17+ game titles (Valorant, CS2, BGMI, League of Legends). Designed and built the entire platform from scratch.

Platform Architecture: Designed event-driven, message-driven microservice architecture (Java J2EE, Next.js) including tournament management engines (single-elimination, double-elimination, round-robin, Swiss), real-time notification and leaderboard systems, streaming services integration, and operations platform.

Platform Durability: The platform continues to power national-scale tournaments in 2026 — five years after initial architecture — demonstrating architectural soundness and long-term impact.

Investor Due Diligence: Led technical due diligence for the \$3M funding round, presenting platform architecture and scalability roadmap to investors.

ASSOCIATE DIRECTOR OF IT

Virtusa — India and USA

May 2022 – March 2023

Pipeline Optimization: Conducted comprehensive CI/CD pipeline analysis resulting in 70% reduction in build and deployment times through advanced DevOps automation practices.

Data Center Migration: Developed and implemented automated data center migration strategy ensuring seamless transitions with minimal downtime.

PRINCIPAL ARCHITECT — CLOUD INFRASTRUCTURE SERVICES

LTI (Larsen & Toubro Infotech) — India

October 2021 – April 2022

Full Stack Digital Operations (FSDO): Architected a comprehensive FSDO offering demonstrating full DevOps capabilities, guiding enterprise clients through platform modernization.

Advisory Services: Provided strategic infrastructure and service modernization advisory to insurance industry clients.

SENIOR ARCHITECT

Cognizant — India, USA (Multiple Clients)

May 2014 – October 2021

DevOps Standardization: Conducted multiple enterprise assessments of DevOps processes, tool standardization, and SDLC practices across telecommunications, pharmaceutical, and payroll sectors.

100% Containerization (ADP): Engineered build and deployment strategy for ADP, successfully containerizing 100% of their HRMS and payroll services.

Hybrid Cloud Migration (Sanofi): Designed and implemented hybrid cloud migration strategy for a leading European pharmaceutical company.

DevOps Metrics Framework (Canadian Telecom): Developed comprehensive DevOps metrics and reporting frameworks delivering actionable performance insights.

ADVISORY IT ARCHITECT

IBM — Australia, India, USA

December 2008 – April 2014

Telstra Platform Architecture: Led architectural support and change management for Telstra's registration and enrollment platform serving millions of Australian consumers.

Regulatory Systems (Idea Cellular): Served as Application Architect for regulatory compliance applications, optimizing system design for telecom compliance requirements.

Integration Architecture: Designed integration architecture for automotive head units connecting entertainment service providers.

EDUCATION

Master of Science in Computer Science — Manipal University, Pune, 2012

Bachelor of Engineering in Computer Science — VNIT (Visvesvaraya National Institute of Technology), Nagpur, 1997

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, SQL

Cloud Platforms: AWS, Google Cloud Platform

DevOps & Infrastructure: Docker, Kubernetes, Jenkins, Ansible, Terraform, CI/CD Pipeline Design

AI/ML: LightGBM, SHAP, Scikit-learn, TensorFlow, Explainable AI (XAI)

Architecture: Microservices, Event-Driven Architecture, Service-Oriented Architecture (SOA)

Performance Engineering: JMeter, Gatling, Load Testing, SLA/SLO Design, Capacity Planning

Databases: Oracle, MongoDB, PostgreSQL, Kafka

INDUSTRIES SERVED (12+ SECTORS)

Banking & Payments: JPMorgan Chase, PayPal, BNY Mellon

Telecommunications: Telstra, Qwest, BT, Cox, Idea Cellular

Pharmaceuticals: Sanofi

Payroll Services: ADP

Media & Entertainment: Shutterstock, The Esports Club

Healthcare: CareFirst

Logistics: JB Hunt

Product Development: Veritas/Symantec

NATIONAL IMPORTANCE CONTEXT

The Consumer Debit Platform at JPMorgan Chase — the largest U.S. bank by assets (\$3.9 trillion) — processes billions of dollars in annual debit transactions across millions of American consumers. U.S. financial services are designated critical infrastructure under Presidential Policy Directive 21. CISA has identified escalating cyber threats against U.S. financial institutions, including a 154% increase in DDOS attacks (2024) and state-sponsored targeting by Russian (GRU) and Iranian actors (CISA Advisories AA25-343A, AA24-241A). The Bureau of Labor Statistics reports 470,000+ unfilled cybersecurity positions nationally, with information security analyst demand growing at 29% — nearly 10x the national average. Mr. Rangari's compensation of \$188,000 exceeds the 90th percentile (\$173,650) for software developers in the Atlanta metropolitan area (BLS OEWS, SOC 15-1252, May 2024), placing him in the top 10% of earners in his occupation and geography.